



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ROHINI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

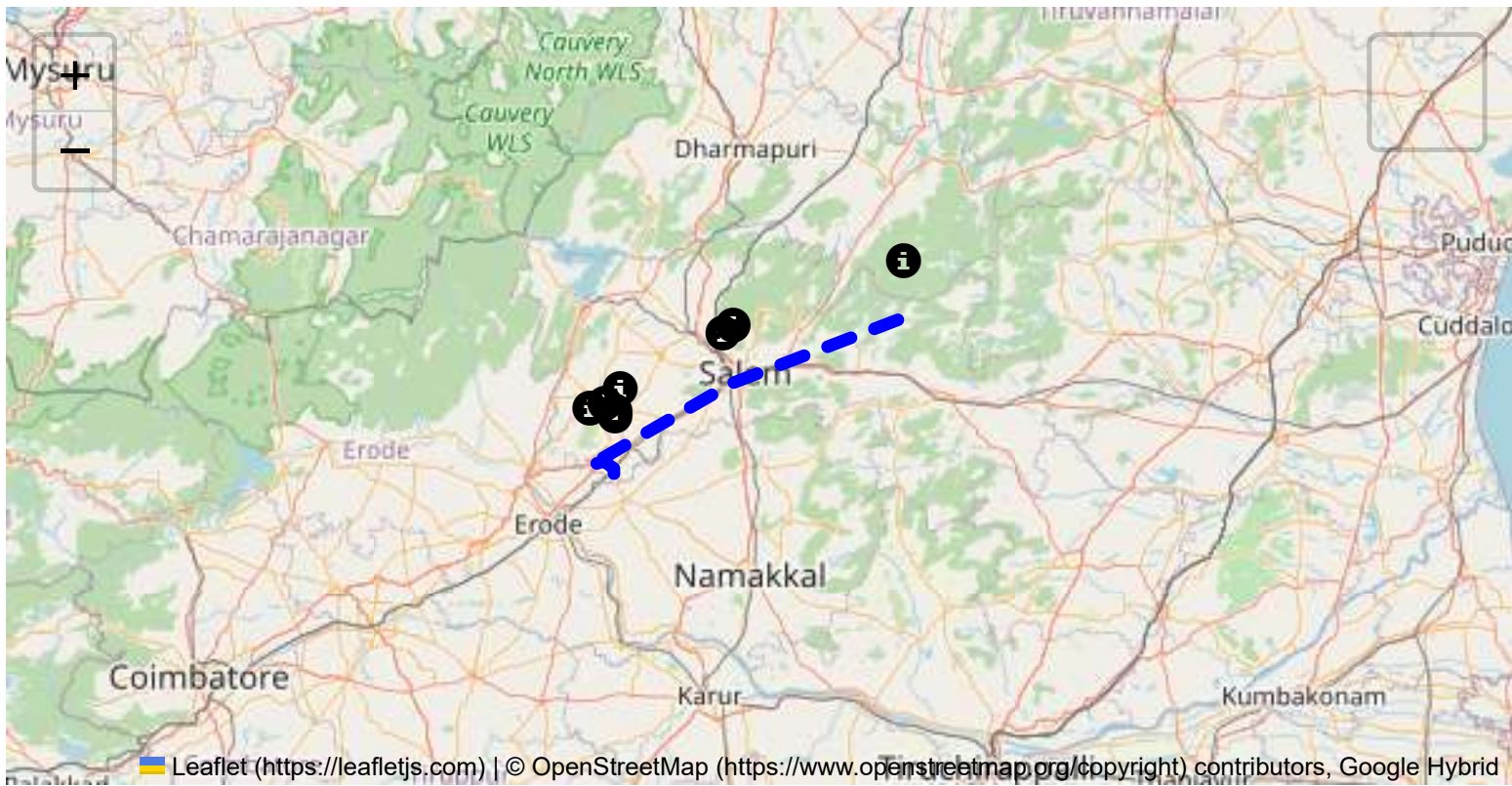
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

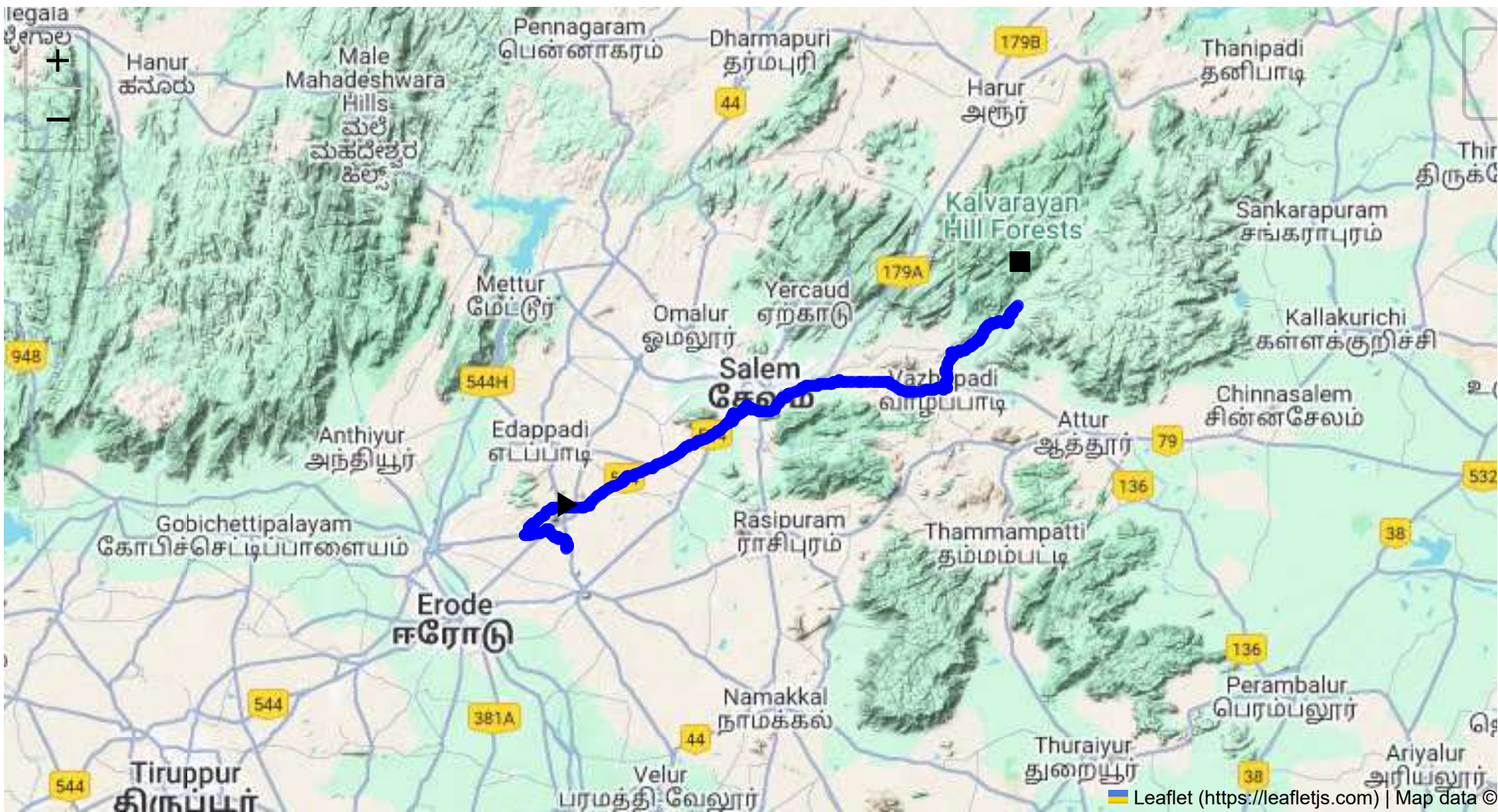
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 103.82 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (11.769919, 78.504776)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 103.82 kilometers, beginning at Sangagiri and ending at Iyyampettai in Tamil Nadu, India. It traverses several major roads including the Sankari RS - Erode Privu Road, Bhavani Main Road, NH544, and Uthamasolapuram in Salem, while passing through Kondalampatti.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** The region experiences a tropical climate with hot summers, a monsoon season from June to September, and cooler winters.
- **Hazards:** Monsoon can cause heavy rain, leading to waterlogging and slippery roads. Visibility may be reduced during foggy winter mornings, and extreme heat can lead to road surface issues and driver fatigue during summer.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is common during morning (7:30 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) rush hours, particularly in urban areas like Salem.
- **Congestion-Prone Areas:** Salem city is a major point of congestion along this route, especially around Uthamasolapuram and Kondalampatti.

4. Assessment of Road Quality and Infrastructure

- **General Road Conditions:** The major highways like NH544 are generally well-maintained, but some rural roads may have potholes or uneven surfaces.
- **Infrastructure:** Adequate signage and lane markings are present on major roads, though less so in rural areas. Bridges and overpasses are routinely maintained.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route:** During heavy congestion or roadblocks, an alternative route through NH44 via Salem, with deviations towards the destination, might be considered. However, this may increase travel distance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Transport of hazardous materials is governed by regulations that dictate specific timing (often nighttime) and routes to reduce risk in populated areas. There may be restrictions on the types of materials and quantities allowed per transport.

7. Overview of Historical Incidents

- **Past Incidents:** There have been incidents of vehicle breakdowns on NH544 due to overloading, alongside sporadic accidents during monsoon due to hydroplaning and visibility issues.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Zones:** The route passes near agricultural zones; any spillage of hazardous materials could have a significant environmental impact.

- **Protected Areas:** While there are no major wildlife sanctuaries directly on this route, local water bodies and farm lands need protection from contamination.

9. Analysis of Communication Coverage

- **Network Coverage:** Generally good in urban areas with some potential dead zones in rural stretches, where mobile signal may be weak.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency services, including fire and police departments, can respond within 20-30 minutes in Salem.
- **Rural Stretches:** Response times may extend to 60 minutes or more due to distance and road conditions.

12. Overall Summary of Risk Assessment

This route, while standard for transport through this region of Tamil Nadu, presents moderate to high risks during monsoon due to weather conditions impacting road safety. Traffic congestion around Salem can cause delays, increasing both transit time and associated risks. Enforcement of strict adherence to hazardous material regulations and preparation for emergency alternate routes are crucial to mitigate risks. Communication coverage is sufficient for most part, but potential dead zones require contingency for communication systems. Environmental sensitivity demands cautious transport to prevent contamination in agricultural areas. Emergency services are reasonably accessible but may require longer response times in rural stretches, underlining the importance of thorough risk assessment and planning for safe passage.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.53560, 77.95773	30 KM/Hr
11	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
12	Turn	High	11.62130, 78.10776	15 KM/Hr
13	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
14	Turn	Medium	11.61224, 78.10469	30 KM/Hr
15	Turn	High	11.63304, 78.12738	15 KM/Hr
16	Turn	Medium	11.62901, 78.16571	30 KM/Hr
17	Turn	Medium	11.65154, 78.35013	30 KM/Hr
18	Turn	Medium	11.65459, 78.37964	30 KM/Hr
19	Turn	High	11.65719, 78.40574	15 KM/Hr
20	Turn	High	11.70455, 78.41564	15 KM/Hr
21	Turn	Medium	11.70556, 78.41348	30 KM/Hr
22	Turn	Medium	11.70578, 78.41337	30 KM/Hr
23	Turn	High	11.70684, 78.41358	15 KM/Hr
24	Turn	Medium	11.71737, 78.44125	30 KM/Hr
25	Turn	Medium	11.72309, 78.44895	30 KM/Hr
26	Turn	Medium	11.72313, 78.44971	30 KM/Hr
27	Turn	Medium	11.72630, 78.45302	30 KM/Hr
28	Turn	Medium	11.72699, 78.45307	30 KM/Hr
29	Turn	Medium	11.74456, 78.46610	30 KM/Hr
30	Turn	Medium	11.74532, 78.48935	30 KM/Hr
31	Turn	Medium	11.74547, 78.48967	30 KM/Hr
32	Turn	Medium	11.75365, 78.49176	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
33	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
34	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium
35	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
37	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
38	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
39	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium
40	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
41	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium
42	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
27	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
29	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
30	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
32	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53560, 77.95773



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61224, 78.10469



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63304, 78.12738



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62901, 78.16571



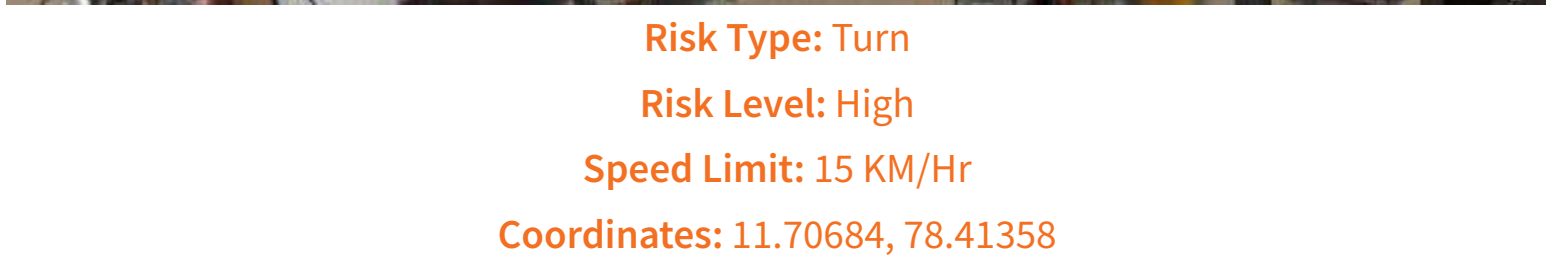
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65154, 78.35013



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65459, 78.37964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65719, 78.40574



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71737, 78.44125



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72309, 78.44895



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72313, 78.44971



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72630, 78.45302



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72699, 78.45307



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74456, 78.46610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74532, 78.48935



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74547, 78.48967



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75365, 78.49176